

As an Executive Director on Nomura's Central Risk Book (CRB) desk, my team is responsible for managing the firm's non-linear exposure, particularly volatility. In the CRB, we often find ourselves naturally short variance—as clients come to us to hedge their equity portfolios, we end up holding the "convexity" that they sell.

To survive and thrive as a Quant Researcher on this desk, you must move beyond Black-Scholes. You need to understand **model-free replication**. This allows us to hedge our volatility books using only vanilla options, without relying on the assumption that volatility is constant or follows a specific stochastic process.

1. The Mathematical Derivation: First Principles

The objective is to replicate the integrated variance $\int_0^T \sigma_t^2 dt$ using a static portfolio of liquid European options.

[A] The Log-Contract Identity

We begin with the fundamental identity derived from Itô's Lemma. For a stock S_t following $dS_t = \mu S_t dt + \sigma_t S_t dW_t$:

1. Define $f(S_t) = \ln(S_t)$.
2. By Itô's Lemma:

$$d(\ln S_t) = \left(\mu - \frac{1}{2} \sigma_t^2 \right) dt + \sigma_t dW_t$$

3. Integrating from 0 to T :

$$\ln \left(\frac{S_T}{S_0} \right) = \int_0^T \left(\mu_t - \frac{1}{2} \sigma_t^2 \right) dt + \int_0^T \sigma_t dW_t$$

4. Under the risk-neutral measure $\mathbb{Q}$, the drift is $\mu_t = r - q$:

$$\mathbb{E}^{\mathbb{Q}} \left[\ln \left(\frac{S_T}{S_0} \right) \right] = (r - q)T - \frac{1}{2} \mathbb{E}^{\mathbb{Q}} \left[\int_0^T \sigma_t^2 dt \right]$$

5. Rearranging, we get the **Variance-Log identity**:

$$\mathbb{E}^{\mathbb{Q}} \left[\int_0^T \sigma_t^2 dt \right] = 2 \left[(r - q)T - \mathbb{E}^{\mathbb{Q}} \left[\ln \left(\frac{S_T}{S_0} \right) \right] \right]$$

[B] The Breeden-Litzenberger Decomposition

We need to replicate the payoff $f(S_T) = \ln(S_T/S_0)$. Any twice-differentiable function $f(S)$ can be expressed as a portfolio of options via the Taylor expansion around a forward F_0 :

1. **Taylor Expansion:**

$$f(S_T) = f(F_0) + f'(F_0)(S_T - F_0) + \int_0^{F_0} f''(K)(K - S_T)^+ dK + \int_{F_0}^{\infty} f''(K)(S_T - K)^+ dK$$

2. **For $f(S) = \ln(S)$:**

- $f'(S) = \frac{1}{S}$
- $f''(S) = -\frac{1}{S^2}$

3. Substituting f'' into the integral:

$$\ln(S_T) = \ln(F_0) + \frac{1}{F_0}(S_T - F_0) - \int_0^{F_0} \frac{1}{K^2}(K - S_T)^+ dK - \int_{F_0}^{\infty} \frac{1}{K^2}(S_T - K)^+ dK$$

- The first two terms represent the forward contract (which has zero cost).
- The integrals represent the **continuum of OTM puts and calls** weighted by $1/K^2$.

[C] Defining Fair Variance K_{var}

Taking the expectation $\mathbb{E}^{\mathbb{Q}}$ of the replication portfolio and multiplying by $2e^{rT}/T$, we arrive at the fair strike:

$$K_{var} = \frac{2e^{rT}}{T} \left(\int_0^{F_0} \frac{P(K)}{K^2} dK + \int_{F_0}^{\infty} \frac{C(K)}{K^2} dK \right)$$

This formula is "model-free"—it requires no assumption about the distribution of S_T , only that there exists a liquid market of options at all strikes K .

2. Application: Theory and Code

In the CRB, we do not have an infinite continuum of strikes. We use a **discretized approximation** (the VIX formula).

```
import numpy as np
import plotly.graph_objects as go

def calculate_variance_swap(strikes, puts, calls, forward, T, r):
    """
    Computes the fair variance strike K_var using the discrete version
    of the Breeden-Litzenberger integral.
    """
    # 1. Calculate step size dK between strikes
    dK = np.diff(strikes)

    # 2. Identify the forward strike index
    idx_f = np.searchsorted(strikes, forward)

    # 3. Summing P(K)/K^2 and C(K)/K^2
    # Note: Using midpoints for the dK summation
    integral_p = np.sum((puts[:idx_f] / strikes[:idx_f]**2) * dK[:idx_f])
    integral_c = np.sum((calls[idx_f:] / strikes[idx_f:]**2) * dK[idx_f:])

    # 4. Applying the VIX formula weighting
    k_var = (2 * np.exp(r * T) / T) * (integral_p + integral_c)

    return k_var
```

```
# Example Plotting Functionality for the ED
def plot_variance_weighting(strikes, options_prices, forward):
    """Visualize the 1/K^2 weighting kernel on the strike continuum."""
    weights = 1.0 / (strikes**2)

    fig = go.Figure()
    fig.add_trace(go.Scatter(x=strikes, y=weights, mode='lines', name='1/K^2
Weighting'))
    fig.add_vline(x=forward, line_dash="dash", line_color="red",
annotation_text="Forward")

    fig.update_layout(
        title="Contribution to Variance Swap Price by Strike",
        xaxis_title="Strike (K)",
        yaxis_title="Weight (1/K^2)",
        template="plotly_white"
    )
    return fig
```

3. Key Follow-Ups & Diagnostic Questions

Question 1: In the discrete sum, how do you handle the interpolation of the smile between existing strike prices?

Answer: The integral assumes a continuum. In reality, we have a sparse grid of strikes. We use **cubic spline interpolation** on the volatility smile (IV vs. Strike) rather than the prices themselves, as the IV surface is much smoother and more stable. We then convert the interpolated IVs back into option prices using the Black-Scholes formula before performing the numerical integration. Failure to do this interpolation creates "kinks" in the kernel, leading to massive errors in K_{var} .

Question 2: The integral theoretically spans from $K = 0$ to $K = \infty$. How do you handle the "truncation" of this integral in production?

Answer: This is a major risk for the CRB. We use **Tail Extrapolation**.

1. For strikes beyond the most OTM liquid options, we assume a flat volatility surface (sticky-strike) to avoid undefined behavior.
2. The weight $1/K^2$ naturally diminishes the impact of far OTM options, but for very low strikes (near $K = 0$), the $1/K^2$ term blows up. We use a **"floor" strike** below which we assume the put price is zero, or use a model-based tail decay. If we don't handle the lower tail correctly, our variance swap price will be artificially inflated, causing us to overpay for protection.